A THREE-PART DEFENSE

THE STRANGE LIFE OF HIGH-DIMENSIONAL DATA.

On Markov populations, the power method & SVD, and a principal-components reading of infrared spectra. A linear-algebra story in three questions.

CINEMATOGRAPHER · Q1
Adithya Kesan Jayakanth

CINEMATOGRAPHER · Q2
K Vishnu Kumar

CINEMATOGRAPHER · Q3
Adithya Anand

A SINGLE MACHINERY

THREE QUESTIONS. ONE EIGENDECOMPOSITION.

Every result we'll defend falls out of an eigendecomposition. We only change the object we decompose.

QUESTION ONE

MARKOV DYNAMICS

A **2×2 column-stochastic** matrix M. Keystone — a permanent residue freezes when the rate is time-varying. **Product, not power.**

QUESTION TWO

POWER METHOD & SVD

Iterate to the dominant eigenvector. Keystone — $X^\top X$ and $XX^\top$ share nonzero spectrum. **Build the smaller square.**

QUESTION THREE

PCA ON IR SPECTRA

3401 × 924 spectra. Keystone — $d_{\text{eff}} \approx 8.7$, 25 clusters, optimal rank-8 via Eckart–Young. **Mask, don't delete.**

QUESTION ONE — MARKOV DYNAMICS

TYPE-1 BECOMES TYPE-2. NOTHING RETURNS.

THE OPERATOR

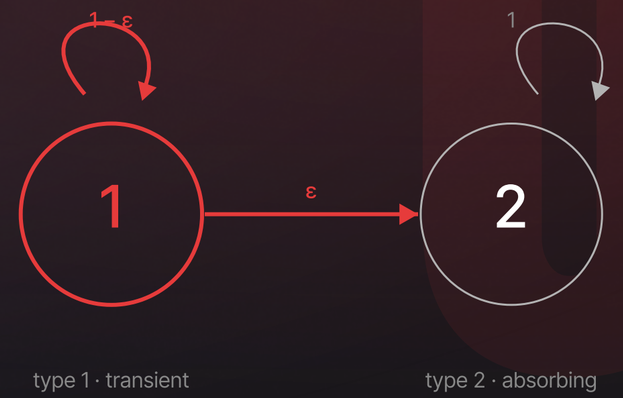
$$P(t{+}1) = M\,P(t), \qquad M = \begin{pmatrix} 1-\varepsilon & 0 \\ \varepsilon & 1 \end{pmatrix}, \quad \varepsilon \in (0,1)$$

CONSERVATION

$$\mathbf{1}^{\top} M = \mathbf{1}^{\top} \implies \mathbf{1}^{\top} P(t{+}1) = \mathbf{1}^{\top} P(t)$$

- **Column-stochastic.** Each column of M sums to 1. $S = p_1 + p_2$ is invariant.
- **Two views, one fact.** Linear algebra ⇔ accounting.

STATE DIAGRAM — FIG. 03.01



Q1 — THE EIGEN-ENGINE

TRIANGULAR. EIGENVALUES READ OFF THE DIAGONAL.

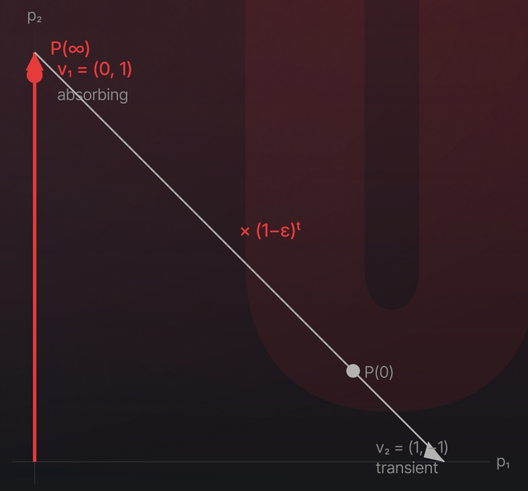
$$M = \begin{pmatrix} 1-\varepsilon & 0 \\ \varepsilon & 1 \end{pmatrix} \Rightarrow \lambda_1 = 1, \;\; \lambda_2 = 1-\varepsilon$$

$$v_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{(absorbing)} \qquad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{(transient)}$$

$$P(0) = c_1 v_1 + c_2 v_2, \quad c_1 = S(0),\ c_2 = p_1(0)$$

- **One line.** $\det(M - \lambda I) = (1-\varepsilon-\lambda)(1-\lambda)$ — roots are the diagonal entries.
- **Why this matters.** In the eigenbasis, M acts component-wise: $Mv_i = \lambda_i v_i$. Every closed-form result downstream is one line.

EIGENVECTORS IN THE (P_1, P_2)-PLANE FIG. 04.01



Q1 — LONG-RUN BEHAVIOR

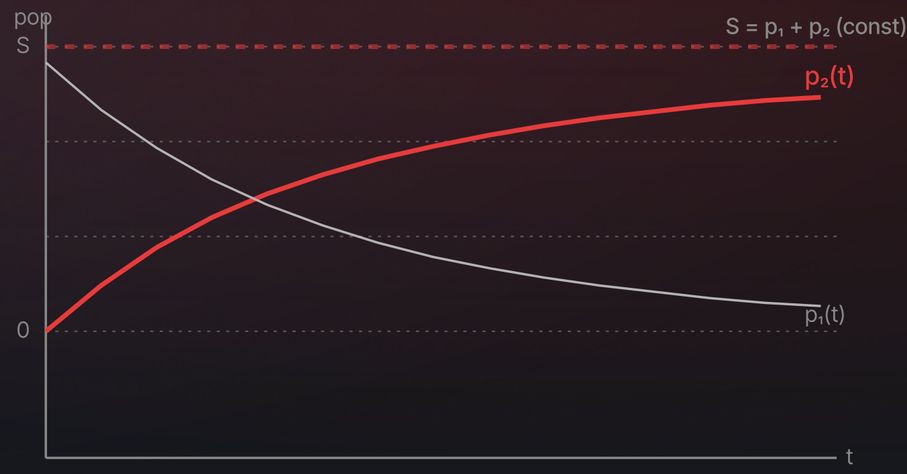
POWERS DIAGONALIZE. TYPE-1 EVAPORATES.

$$P(t) = c_1\, v_1 + c_2\, (1-\varepsilon)^t\, v_2$$

- **Diagonalize and pull out powers.** $M^t v_i = \lambda_i^t v_i$.
- **Geometric decay.** $(1-\varepsilon)^t \to 0$ — the second term dies.
- **Absorbing limit.** $\lim_{t\to\infty} P(t) = c_1 v_1 = (0,\ S(0))$. All mass becomes type-2.
- **Numerical check (Q1.f).** Iterate vs. closed form agree to 10^{-16} — the floating-point floor.

DECAY RATE	LIMIT · TYPE-1	ITER VS CLOSED
1-ε	0	10

POPULATIONS · E = 0.15 — FIG. 05.01



THE Q1 KEYSTONE

PRODUCT, NOT POWER.

The decay rate is now time-varying: $\varepsilon(t) = \varepsilon_0\, e^{-t/t_0}$. Each matrix $M(t)$ differs — and yet, crucially, every $M(s)$ shares the same eigenvectors v_1, v_2.

$$P(t) = c_1 v_1 + c_2 \left[\prod_{s=0}^{t-1} (1 - \varepsilon(s))\right] v_2$$

Because $\sum_s \varepsilon(s) < \infty$, the infinite product converges to a strictly positive limit $\Pi_\infty > 0$. **A power decays. A product freezes.** A permanent residue of type-1 — measured in our numerics at $\approx 10^{-3}$ — never disappears.

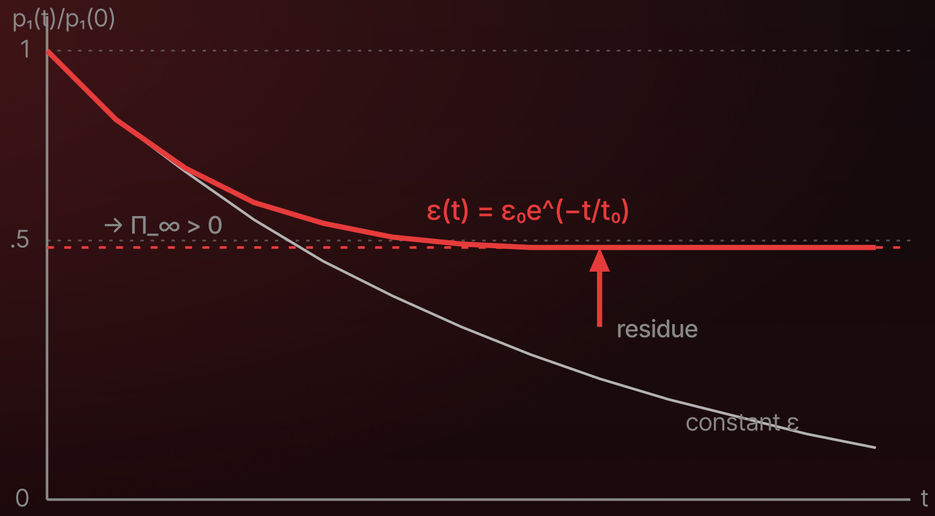
RESIDUE · Π∞

~10 −3

ITER VS CLOSED

10 −12

P₁(T) / P₁(0) FIG. 06.01



QUESTION TWO — POWER METHOD & SVD

REPEATED MULTIPLICATION AMPLIFIES THE DOMINANT MODE.

$$x_{k+1} = \frac{X x_k}{\|X x_k\|}, \qquad X = X^\top,\ |\lambda_1| > |\lambda_2| \geq \cdots$$

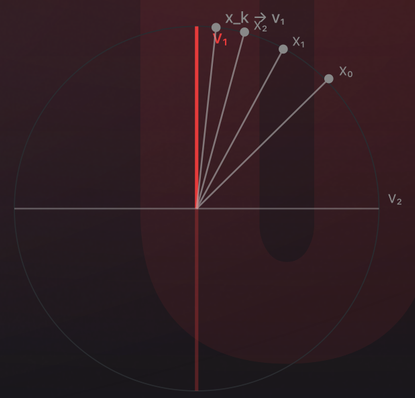
WHY IT WORKS

$$X^k x_0 = \lambda_1^k \left[\alpha_1 v_1 + \textstyle\sum_{i>1} \alpha_i (\lambda_i/\lambda_1)^k v_i\right]$$

- **Every other term shrinks.** $(\lambda_i/\lambda_1)^k \to 0$ for $i > 1$.
- **After normalization,** $x_k \to \pm v_1$ — Rayleigh: $\lambda_1 = x_k^\top X x_k$.
- **Sign is undetermined.** Eigenvectors are equivalence classes mod sign.

ITERATES ALIGNING WITH V_1

FIG. 07.01 · $|\Lambda_2/\Lambda_1| = 0.5$



Q2 — CONVERGENCE

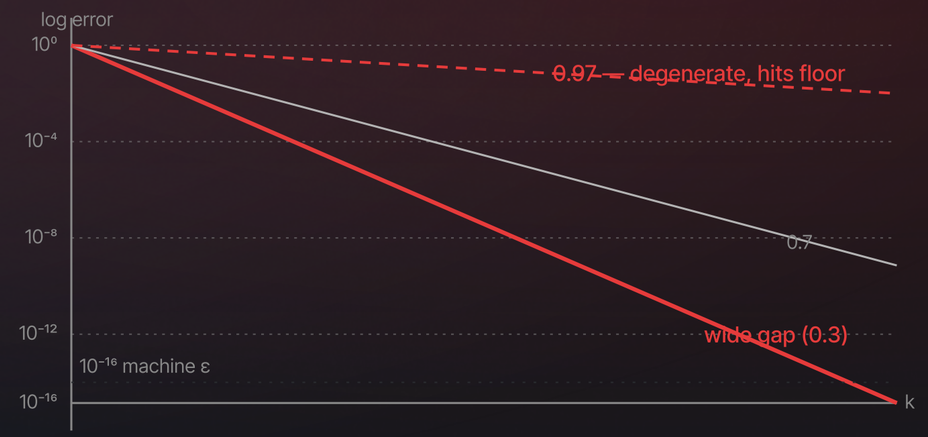
WIDE GAP = FAST. CLOSED GAP = FAILURE.

$$\|x_k - v_1\| = \mathcal{O}\big(|\lambda_2/\lambda_1|^k\big)$$

- **Iterations to ε-accuracy** $\approx \log(1/\varepsilon)/\log(\lambda_1/\lambda_2)$.
- **Pathological case.** $X = \mathrm{diag}(1, -1)$ — iterate oscillates forever.
- **Near-degenerate** iter count $\sim 1/(1 - |\lambda_2/\lambda_1|)$ — per-step separation drowns in the 10^{-16} rounding floor.
- **Verdict.** Spectral gap is a number — but also a verdict on whether the iteration works at all.

LOG‖X_K − V₁‖ VS K

FIG. 08.01



THE Q2 KEYSTONE

TWO MATRICES. ONE SPECTRUM.

$$X^\top X\, v = \lambda v,\ \lambda \neq 0 \implies XX^\top (Xv) = \lambda (Xv)$$

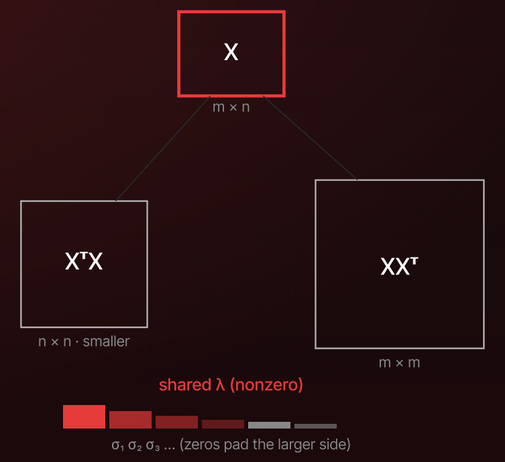
$$\text{PSD:}\quad w^\top (X^\top X) w = \|Xw\|^2 \geq 0$$

$$\sigma_i = \sqrt{\lambda_i (X^\top X)} = \sqrt{\lambda_i (XX^\top)}$$

- **Practical lesson.** Diagonalize the smaller square — size $\min(N, M)$. Same answer, far cheaper. We use this directly in Q3(f).
- **Worked example.** $X = [[1,0],[0,1],[1,1]] \to \lambda = 1, 3$; $\sigma_1 = \sqrt{3}$, $v_1 = (1,1)/\sqrt{2}$, $u_1 = (1,1,2)/\sqrt{6}$.

SHARED SPECTRUM, TWO ROUTES

FIG. 09.01 · $\Sigma_i = \sqrt{\Lambda_i}$



QUESTION THREE — PCA ON IR SPECTRA

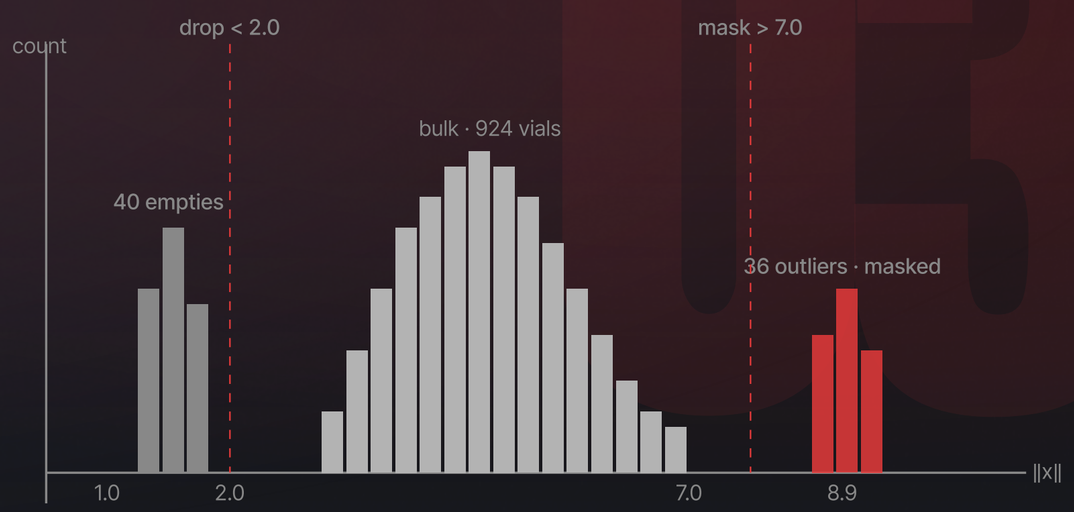
TWO GAPS IN THE NORM. MASK, DON'T DELETE.

- **File shape: 3401 × 1000**. Prompt said $M{=}3041$ — digit transposition. We used the true 3401.
- **Two natural gaps.** 40 empties below ≈ 1.6, 36 outliers above ≈ 8.9.
- **Drop the empties; mask the outliers.** Non-destructive — the outliers form one isolated cluster.
- **Working size:** $\tilde{N} = 924,\ M = 3401.$

$$C = \tfrac{1}{\tilde{N}}\,\tilde{X}^{\top}\tilde{X}, \quad \operatorname{tr} C = \textstyle\sum_j \mathrm{Var}_j = \sum_i \lambda_i$$

TR C	MOST-VAR FEAT.	WORKING N
7.90	i_▲_ q10	924

HISTOGRAM OF ROW-NORMS $\|X_i\|$ — FIG. 10.01

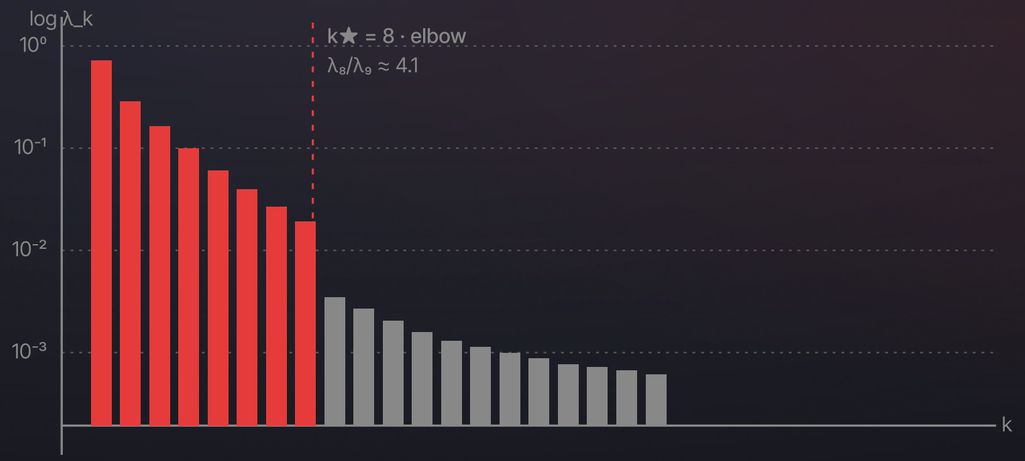


Q 3 — INTRINSIC DIMENSION

ELBOWS LIE. PARTICIPATION RATIO DOESN'T.

LOG Λ_K SPECTRUM

FIG. 11.01 · $\Lambda_1 \approx 31\%$ OF TR C



$$d_{\text{eff}} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}$$

- **$\lambda_1/\lambda_2 \approx 3.5$.** Strong leading direction.
- **Elbow alone is fragile**: bootstrap moves k between 6 and 11.
- **"95% rule" misleads** — would demand 667 components.
- **d_{eff} is bootstrap-stable** — no hard cut to pick.

ELBOW	D_EFF	CI · BOOTSTRAP
k★ 8	8.7	[7.6. 10]

Q3 — WHAT V_1 MEANS

V_1 IS A CONTRAST, NOT OVERALL BRIGHTNESS.

$$\cos(v_1,\ \bar{x}_{\text{centred}}) \approx 0.92$$

- **Two narrow IR bands** carry the leading signal.
- **~27% negative loadings** — samples differ in the RATIO of band A to band B.
- **Scale-free.** Cosine ignores absolute brightness; a constant offset cannot fake this.

% NEGATIVE	LEADING VAR
27 %	91

LOADINGS OF V_1 ACROSS 3401 WAVENUMBERS — FIG. 12.01

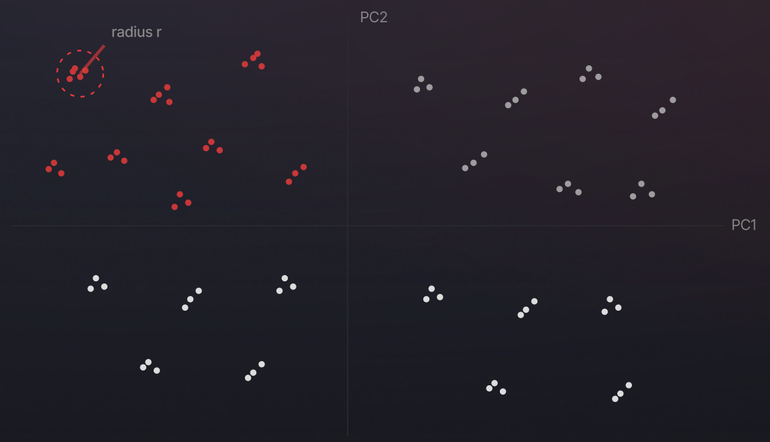


Q 3 — TWENTY-FIVE

TOP-4 SCORES. LEADER. 25 CLUSTERS. GRAM AGREES.

PC1 × PC2 · 924 SAMPLES

FIG. 13.01 · STABLE R ∈ [6, 12] × NN



$$Z = \tilde{X} V_4 \in \mathbb{R}^{924 \times 4} \xrightarrow{\text{leader}(r)} 25 \text{ clusters}$$

THE GRAM SHORTCUT

$$G = \frac{1}{\tilde{N}} \tilde{X} \tilde{X}^\top \in \mathbb{R}^{924 \times 924}$$

- **Since** $\tilde{N} \ll M$, diagonalizing G costs $(M/\tilde{N})^2 \approx 13.5\times$ less.
- **Top-8 eigenvalues** match C's to 10^{-6}.
- **Must agree.** SVD identity: $Z = U\Sigma$ either way — sign per axis is the only freedom.

THE Q3 KEYSTONE

TOP-8 IS OPTIMAL. ECKART-YOUNG SAYS SO.

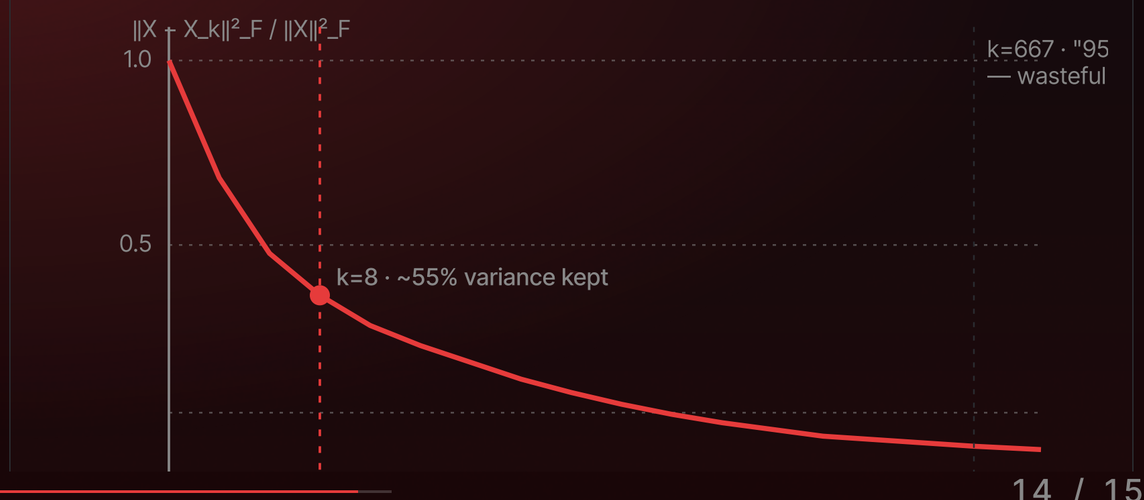
$$\underbrace{24 \times 37}_{\text{replicates}} + \underbrace{36}_{\text{outliers}} = 924$$

- **25 unique food samples** — 24 tight clusters of ~37 replicates + 1 isolated cluster.
- **Intrinsic dim ≈ 8** ≈ the ten dominant molecular signatures expected from IR.

ECKART–YOUNG

$$\min_{\operatorname{rank} B = k} \|\tilde{X} - B\|_F^2 = \sum_{i>k} \sigma_i^2$$

FROBENIUS ERROR VS RANK K FIG. 14.01 · K★ = 8



A FORWARD LOOK

A PRINCIPLED NOISE FLOOR. FOR THE ORAL DEFENSE.

- **What the report asserts.** A "multiplicative noise boundary" justifies the elbow at $k = 8$.
- **What we never derived.** A noise model that says where the bulk SHOULD end.
- **The fix.** Even pure noise has a predictable eigenvalue band. Marchenko–Pastur gives the closed-form edge.

$$\lambda_{\pm} = \sigma^2 \left(1 \pm \sqrt{\gamma}\right)^2, \quad \gamma = M/N$$

- **Our data:** $\gamma = 3401/924 \approx 3.68$. With $\hat{\sigma}$ from masked bulk, λ_+ falls between λ_8 and λ_9.
- **What it buys us.** An adversary-proof reason to cut at 8 — not by inspection.

DENSITY · EMPIRICAL VS MP FIG. 15.01 · EDGE = Λ_+

